

The turbulent boundary layer as an unconfined intermittent canopy:

A structural synthesis

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Abstract

We develop a structural interpretation of the outer region of a zero-pressure-gradient turbulent boundary layer by combining three results that are usually discussed separately: the hierarchy of wall-attached momentum-transfer and vortex structures, the temporal similarity of those structures, and the geometry and dynamics of the turbulent/non-turbulent interface. The evidence supports an *intermittent-canopy* picture in which self-similar motions of many sizes overlap below the boundary-layer edge, while only the largest motions reach the intermittent region. Intense attached Reynolds-stress objects in channels have dimensions approximately proportional to their height and, at the identification threshold used in the source study, carry about 60% of the Reynolds stress while occupying less than 8% of a wall-parallel plane. Related vortex clusters have comparable aspect ratios and a scale-dependent population. Independent boundary-layer experiments and DNS identify analogous wall-attached packet and velocity structures through the logarithmic layer, although not at the T/NT interface itself. Time-resolved tracking shows that tall attached structures preserve this similarity in time: their lifetimes scale with their height and u_τ . They can therefore contribute to a canopy without introducing a new outer time scale that would break wall similarity. At the outer boundary, we adopt the interface definition and conditional coordinate established by [Borrell and Jiménez \[2016\]](#). Their data show that turbulent-level vorticity and strain terminate near the edge: strain drops by more than one order of magnitude before the much weaker potential-flow branch decays exponentially. On the turbulent side, strain-normalised stretching and strain-eigenvalue statistics remain similar between the interface buffer and the core. We hypothesise that the interface is the convoluted outer envelope at which the wall-attached cluster hierarchy terminates: the intermittent region is where that hierarchy becomes sparse. The resulting cluster-to-interface connection is deliberately weak: attached structures are established in the cluster-bearing core, and interface-conditioned data show similar normalised physics up to the local turbulent edge, including its intermittent excursions, but the two have not been labelled jointly. This weak connection yields direct predictions for joint cluster/interface measurements.

Keywords: wall-bounded turbulence; turbulent boundary layer; coherent structures; attached eddies; turbulent/non-turbulent interface; intermittency

1 Introduction

Descriptions of zero-pressure-gradient turbulent boundary layers (ZPG TBLs) commonly use several complementary languages. Mean-flow descriptions distinguish viscous, logarithmic, wake and intermittent regions. Structural descriptions invoke streaks, sweeps, ejections, vortex clusters and attached eddies. Interface descriptions focus on the boundary separating rotational turbulence from the irrotational free stream [[Corrsin and Kistler, 1955](#), [Borrell and Jiménez, 2016](#)]. These descriptions refer to different observables and cannot be identified with one another without an explicit argument.

Planar particle-image velocimetry established that the outer TBL contains repeated combinations of spanwise vorticity and strong Q2 motion [[Adrian et al., 2000](#)]. Spanwise scales grow

linearly with wall distance through the logarithmic layer [Tomkins and Adrian, 2003], three-dimensional experiments recover vortex-packet signatures in an actual ZPG boundary layer [Dennis and Nickels, 2011], and boundary-layer DNS identifies self-similar wall-attached velocity structures up to approximately 0.18δ [Hwang and Sung, 2018]. Channel DNS provides the complementary gradient-based result that intense Reynolds-stress regions and vortex clusters include wall-attached, approximately self-similar families [del Álamo et al., 2006, Lozano-Durán et al., 2012]. Time-resolved DNS showed that the geometrical similarity of tall attached structures extends to their lifetimes [Lozano-Durán and Jiménez, 2014]. In parallel, conditional analysis of the turbulent/non-turbulent (T/NT) interface showed a sharp separation between turbulent-level gradients and a much weaker potential-flow strain branch [Borrell and Jiménez, 2016].

We ask whether these observations can be organised into one economical picture. The proposed *intermittent-canopy hypothesis* is:

The logarithmic and outer regions contain a hierarchy of overlapping, approximately self-similar turbulent structures. Their population and spatial coverage decrease with height. Near the boundary-layer edge, the union of the surviving structures becomes sparse, and its convoluted vortical boundary is observed as the T/NT interface.

The term *canopy* conveys overlap, a distribution of heights and an irregular upper envelope. It does not imply a forest of isolated, permanent vortex tubes rooted at the wall. The structures extracted from DNS are multiscale regions rather than idealised individual vortices. In addition, most three-dimensional object evidence discussed below was obtained in channels, whereas the interface evidence was obtained in a spatially developing boundary layer. The synthesis is plausible only to the extent that the relevant logarithmic-layer organisation is shared by the two flows.

This article is a synthesis rather than a report of a new simulation. We consequently distinguish (i) observations displayed directly by published statistics, (ii) inferences supported by those observations, and (iii) the untested connection between the attached hierarchy and the interface. That distinction prevents a wall-attached Reynolds-stress object, a connected vortex cluster and a region enclosed by a T/NT interface from being treated as interchangeable.

2 Definitions and scales

Let x , y and z denote the streamwise, wall-normal and spanwise directions. The velocity fluctuations are (u, v, w) , the friction velocity is u_τ , the kinematic viscosity is ν , and δ_{99} is the height at which the mean velocity reaches 99% of its free-stream value. Wall units are denoted by a superscript $+$, so that

$$\delta_{99}^+ = \frac{u_\tau \delta_{99}}{\nu}. \quad (1)$$

A quadrant object is a connected region in which the instantaneous tangential Reynolds stress exceeds a prescribed threshold. Ejections are Q2 events ($u < 0$, $v > 0$), and sweeps are Q4 events ($u > 0$, $v < 0$). In the object analysis of Lozano-Durán et al. [2012], an object is termed attached when its lowest point lies within approximately 20 wall units of the wall. This is an operational definition of geometric connectivity, not a statement about the birthplace of the full object.

Let $\omega = |\omega|$ be the vorticity magnitude. We adopt without modification the vorticity-interface convention and interface-conditioned coordinate established by Borrell and Jiménez [2016]. Their dimensionless interface scale is

$$\omega^* = \omega^+ \sqrt{\delta_{99}^+} = \omega \frac{\nu}{u_\tau^2} \sqrt{\frac{u_\tau \delta_{99}}{\nu}}. \quad (2)$$

The corresponding dimensional edge-vorticity scale is

$$\omega_e \sim \frac{u_\tau^2}{\nu \sqrt{\delta_{99}^+}} = \sqrt{\frac{u_\tau^3}{\nu \delta_{99}^+}}, \quad (3)$$

up to constants. It follows from the local-equilibrium estimate $\nu \omega_e^2 \sim u_\tau^3 / \delta_{99}$. Note that $\sqrt{\delta_{99}^+}$, rather than its inverse, appears in the definition of ω^* .

With the interface convention fixed as in [Borrell and Jiménez \[2016\]](#), let

$$\gamma(y) = \Pr\{\text{point at height } y \text{ lies on the turbulent side}\}. \quad (4)$$

This is not generally equal to $\Pr\{h_k > y\}$, where h_k is a cluster height. Pointwise coverage also depends on each object's footprint, overlap and connectivity. A cluster-height cumulative distribution is therefore not, by itself, an intermittency profile.

3 A self-similar structural hierarchy

3.1 Three-dimensional attached objects

The vortex-cluster analysis of [del Álamo et al. \[2006\]](#) separates compact detached objects from tall attached objects. Rather than using one arbitrarily selected cluster as evidence for the hierarchy, figure 1 shows the population statistics directly. The Cartesian bounding dimensions scale approximately as

$$\Delta_x \simeq 3\Delta_y, \quad \Delta_z \simeq 1.5\Delta_y. \quad (5)$$

The data in figure 1 are from the S1900 channel at $Re_\tau \simeq 1900$, comparable to the upper friction-Reynolds-number range of the boundary-layer interface data discussed in section 4. Reynolds number is therefore not the main mismatch between the two datasets; flow geometry and object definition are. Figure 1 supplies the wall-similar cluster hierarchy required by the hypothesis; the interface side of the proposed identification is developed in section 4.

The number density per unit wall area and height has a power-law tail. As the identification threshold approaches the percolation transition, its exponent approaches -3 , as expected for a Townsend-like geometrically self-similar population [[Townsend, 1976](#), [del Álamo et al., 2006](#)].

[Lozano-Durán et al. \[2012\]](#) performed a corresponding three-dimensional analysis of intense momentum-transfer events. Their objects separate into attached and detached populations (figure 2). At their selected threshold, attached Q2 and Q4 objects occupy less than 8% of a wall-parallel plane but account for approximately 60% of the mean Reynolds shear stress at all heights. This threshold-specific statement is more defensible than claiming that such objects fill the flow or carry nearly all of the stress.

The attached Q family has dimensions

$$\Delta_x \simeq 3\Delta_y, \quad \Delta_z \simeq \Delta_y, \quad (6)$$

except for the largest objects crossing the channel centreline. Figure 3 displays the distributions underlying equation (6). The similarity between equations (5) and (6) is significant, but it does not imply that the vortex and momentum-transfer objects are identical. More than 80% of attached vortex clusters intersect a Q object, whereas the reverse probability decreases with height because tall Q objects outnumber tall clusters [[Lozano-Durán et al., 2012](#)].

The preferred arrangement is a side-by-side Q2–Q4 pair with a vortex cluster associated with the Q2 and with the shear layer below the Q4. Figure 4 compares the conditional geometry with one instantaneous realisation. The instantaneous field is appreciably less regular than its conditional average.

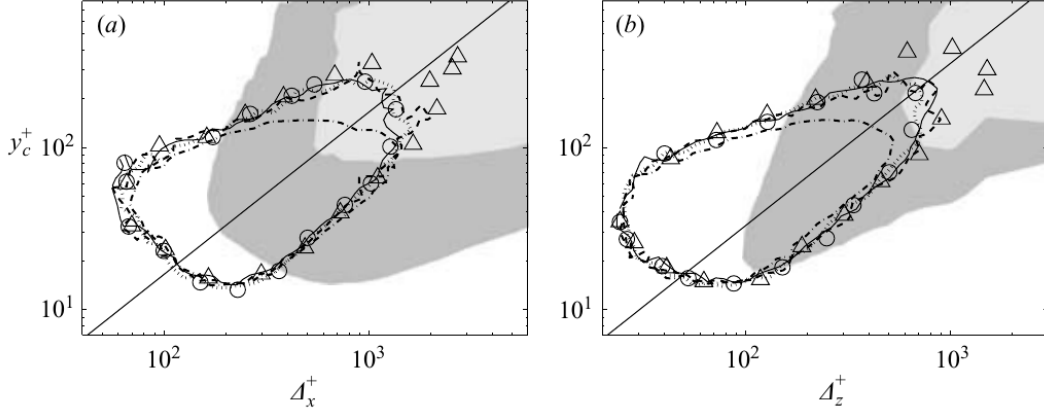


Figure 1: Joint distributions of attached-cluster streamwise (a) and spanwise (b) dimensions versus the wall distance of their centres. The diagonal trends are the direct evidence for geometrical self-similarity. Data are from the S1900 channel at $Re_\tau \simeq 1900$. Reproduced from figure 7 of del Álamo et al. [2006].

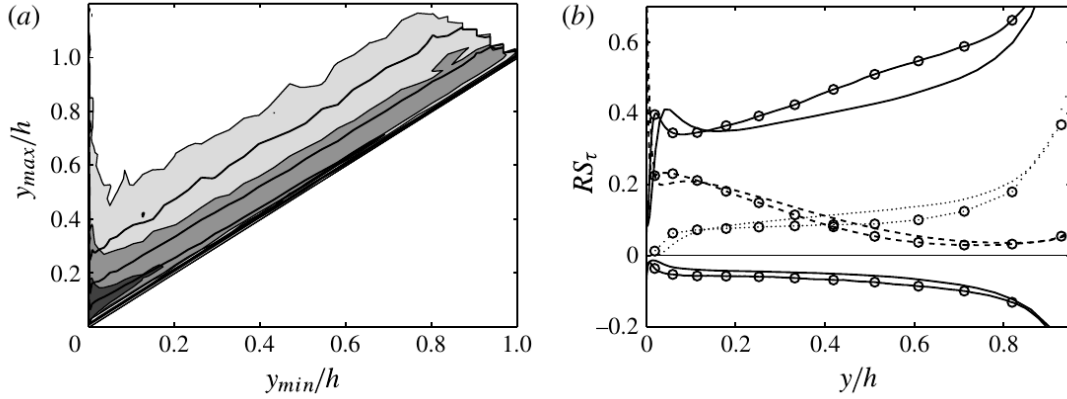


Figure 2: Left: joint distribution of the lowest and highest wall distances of identified Q objects. Right: fractional contributions to Reynolds stress from attached Q2 and Q4 objects, detached objects and counter-gradient events. Reproduced from figure 2 of Lozano-Durán et al. [2012].

The channel result is not the only evidence for an attached hierarchy. In ZPG boundary layers, Tomkins and Adrian [2003] found spanwise length scales growing linearly with wall distance, Dennis and Nickels [2011] measured three-dimensional vortex-packet signatures, and Hwang and Sung [2018] identified self-similar attached structures in all three velocity components. The last study reconstructed logarithmic intensity from the attached population over $100 < y^+ < 0.18\delta^+$. These studies concern velocity or planar vortex signatures rather than the connected gradient clusters used here, and they stop below the intermittent edge. Nevertheless, they anchor the wall-attached hierarchy in a boundary layer at approximately 0.2δ , rather than only in channels.

Together, these observations provide three ingredients of the canopy interpretation: objects span a hierarchy of heights; their wall-parallel dimensions grow with height; and sparse intense objects make a disproportionate contribution to momentum transfer. They do not make velocity, stress and vorticity objects interchangeable.

3.2 The largest objects are composite

Self-similarity breaks down for the largest channel structures. Objects crossing the centreline account for roughly one third of the stress carried by attached Q objects. Fewer than 1% of

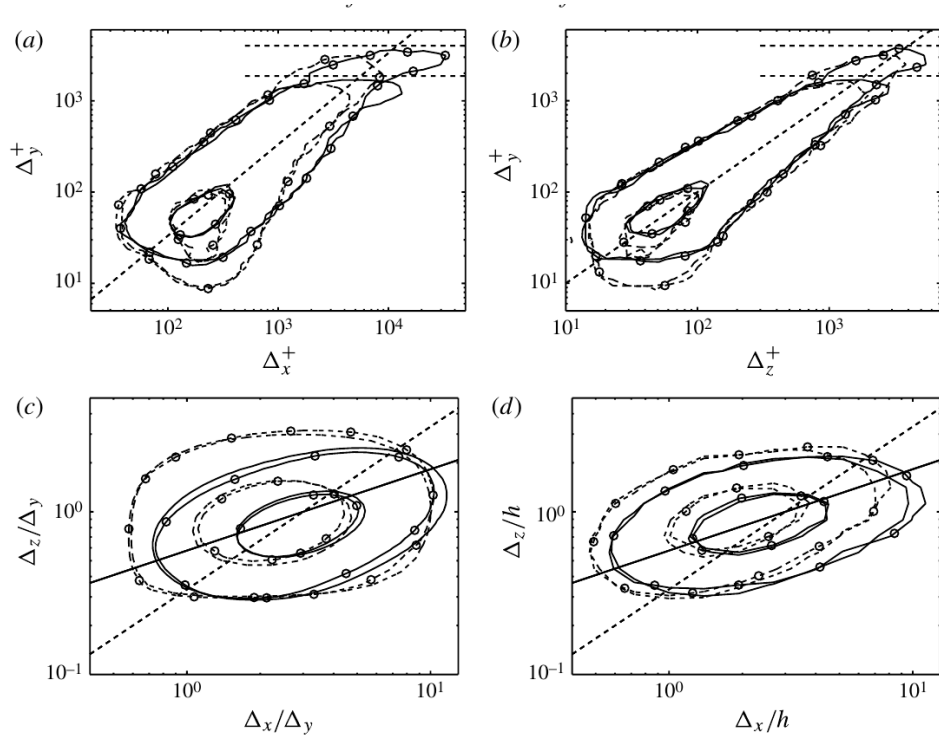


Figure 3: Joint distributions of the dimensions of attached Q2 and Q4 objects. The diagonal trends in panels (a,b) support dimensions proportional to object height. The largest objects in panel (d) depart from the simpler self-similar family. Reproduced from figure 5 of [Lozano-Durán et al. \[2012\]](#).

all identified objects account for about 60% of the volume of the attached population, and streamwise lengths can approach 20δ [[Lozano-Durán et al., 2012](#)]. Figure 5 shows that a very-large Q2 is visibly assembled from smaller pieces.

Consequently, the height of a percolated object is not necessarily the size of one dynamical eddy, and a long correlation or spectral mode need not represent one continuous vortex packet. Spectral evidence likewise contains multiple geometrical regimes: the width of the streamwise-velocity spectrum varies approximately as the square root of its length, and the largest global modes require an outer velocity scale [[del Álamo et al., 2004](#)]. Any canopy model must distinguish active momentum-transfer motions from longer, partly inactive streamwise-velocity organisation.

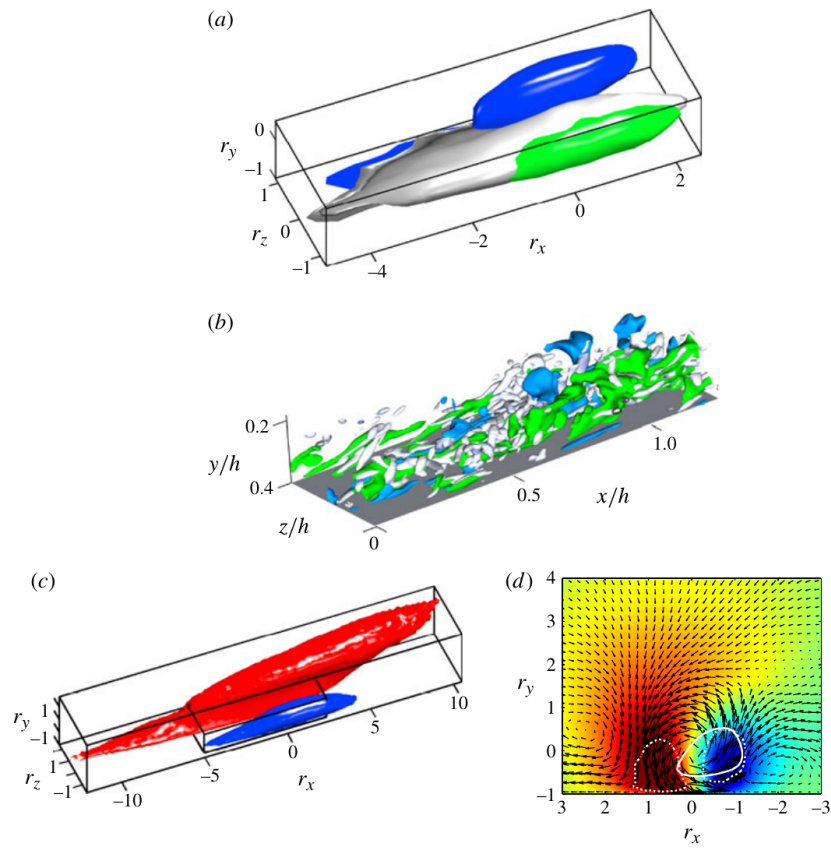


Figure 4: Conditional geometry, an instantaneous realisation, and the associated velocity field of a Q2–Q4 pair. Green denotes Q2, blue Q4 and grey the vortex cluster in the upper panels. Reproduced from figure 12 of [Lozano-Durán et al. \[2012\]](#).

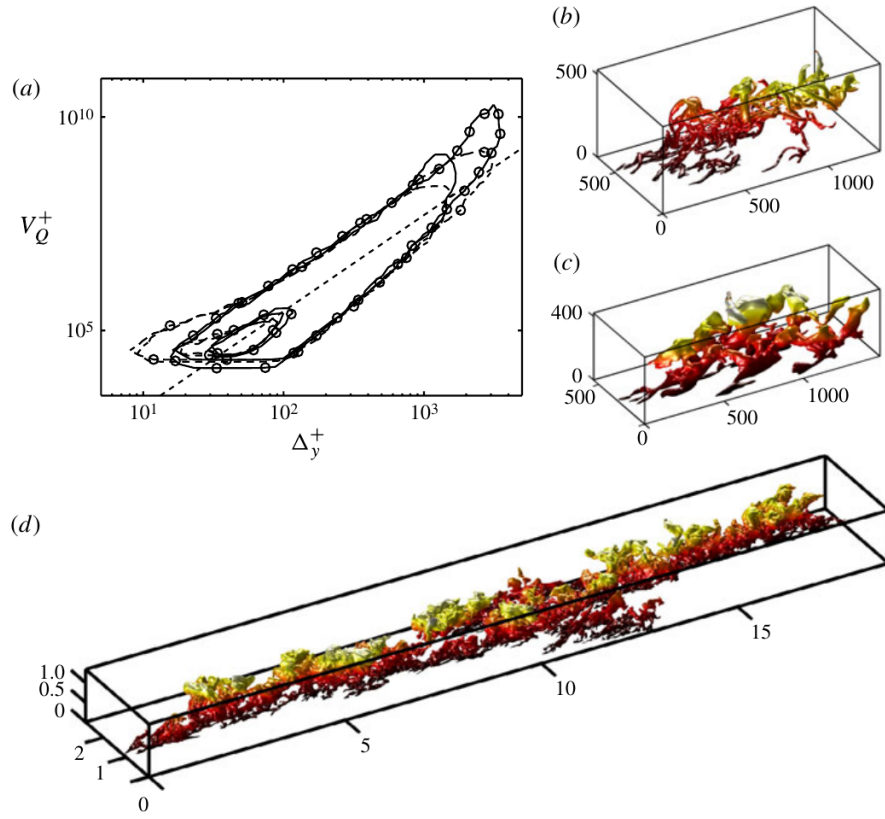


Figure 5: An attached vortex cluster (b), an attached Q2 (c) and a very-large centreline-crossing Q2 (d). The global object is visibly composite. Reproduced from figure 8 of [Lozano-Durán et al. \[2012\]](#).

3.3 Preservation of wall similarity

The time-resolved result needed by the cluster-to-interface hypothesis is whether the structures forming the canopy preserve wall similarity. [Lozano-Durán and Jiménez \[2014\]](#) found that tracked tall attached branches remain geometrically self-similar and that their lifetimes scale as

$$\frac{Tu_\tau}{\Delta_y} = O(1). \quad (7)$$

The tracked structures therefore introduce neither a separate outer length nor a separate outer time scale: their dimensions scale with Δ_y , and their duration with Δ_y/u_τ . This is the role of the time-resolved evidence in the hypothesis: the coherent structures can form a canopy while preserving wall similarity. The identification of that canopy with the boundary-layer interface is then supplied by the termination and inner-side evidence in section 4.

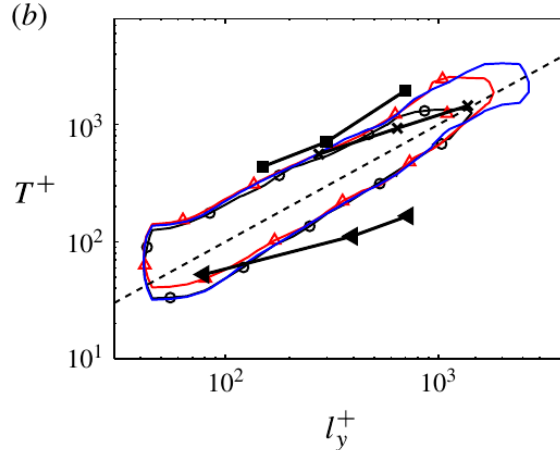


Figure 6: Lifetime of tall attached Q branches versus branch height. The dashed diagonal is $T^+ = l_y^+$. The temporal scale is set by the same height and friction velocity that describe the attached hierarchy. Cropped from figure 9(b) of [Lozano-Durán and Jiménez \[2014\]](#).

4 Termination at the established T/NT interface

We use the interface-conditioned statistics reported by [Borrell and Jiménez \[2016\]](#) directly. Because their coordinate follows the local interface, the conditional ensemble includes both valleys below and bulges above the mean δ_{99} . The present question is what those statistics imply for the canopy framework.

4.1 Separation of turbulent and potential strain

Figure 7 must not be read as showing turbulent strain extending continuously into the free stream. The dominant turbulent strain and vorticity occupy comparable $O(1)$ star-unit levels below the edge. Across the edge, the strain support drops by more than one order of magnitude. Only after that drop does the much weaker irrotational branch follow the exponential decay associated with a wall-parallel wavelength of order $2\delta_{99}$.

For the canopy hypothesis, the important observation is the termination of the high-amplitude rotational and strain field. The weak potential branch outside is a non-vortical response of the free stream, not a continuation of the turbulent canopy.

4.2 Core-like organisation up to the turbulent side

Figure 8 concerns the turbulent side of the interface, not continuity into the potential flow. Its key conclusion is that vorticity in the interface buffer lives in essentially the same straining environment as the turbulent core. Panel (a) follows the joint vorticity and strain distributions towards the interface, and panel (b) follows vortex stretching and compression. In panels (c,d), the samples at 7η and 100η correspond respectively to the interface buffer and the turbulent core. Once normalised by the local strain magnitude, the stretching/compression and strain-eigenvalue distributions are similar. The internal organisation therefore remains core-like as the cluster hierarchy approaches its outer termination, even though the amplitudes subsequently collapse across the interface.

This is the weak bridge used in the present hypothesis. Independent boundary-layer evidence places a wall-attached hierarchy in the cluster-bearing flow up to approximately $0.2\delta_{99}$, while the interface-conditioned statistics show essentially the same normalised straining physics from the core to the local turbulent edge near δ_{99} , including the ensemble of peaks above it. We infer—but do not directly observe—that the attached cluster hierarchy carries that physics to

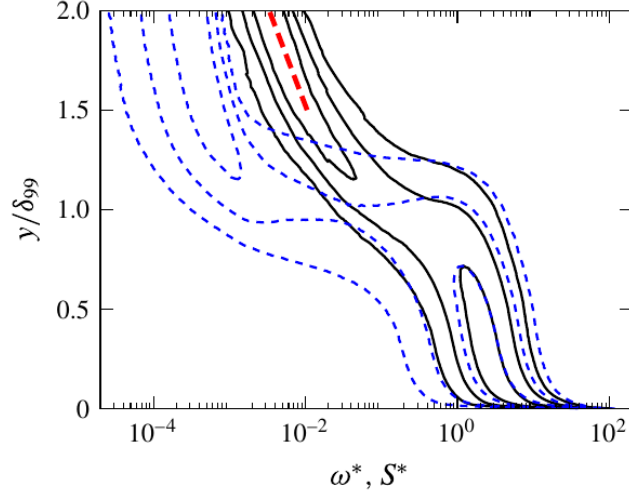


Figure 7: Joint wall-normal distributions of vorticity (blue dashed contours) and strain (black solid contours). Turbulent-level strain drops by more than one order of magnitude near the edge; the remaining weak potential branch follows the exponential decay marked in red. This is a separation of regimes, not continuous turbulent strain into the free stream. Reproduced from figure 20 of [Borrell and Jiménez \[2016\]](#).

the edge and terminates there. The T/NT interface is proposed to be the envelope of those terminations.

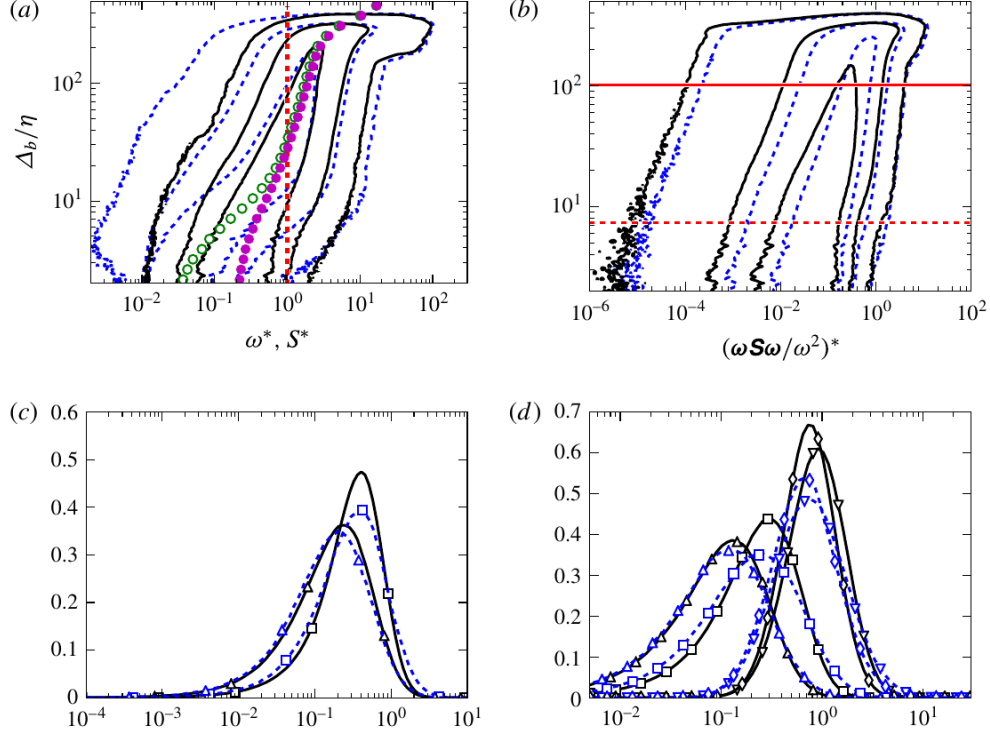


Figure 8: Conditional gradient statistics on the turbulent side of the vorticity interface. Blue dashed curves sample the interface buffer at 7η ; black solid curves sample the core at 100η . Their near collapse after local-strain normalisation shows persistence up to the edge, not continuity across it. Reproduced from figure 23 of [Borrell and Jiménez \[2016\]](#).

5 A weak cluster-to-interface connection

The argument deliberately combines strong observations through one weak inference.

1. In channels, attached vortex and momentum-transfer clusters form a wall-similar hierarchy, are associated with ejections, and preserve height-based length, velocity and time scaling to the outermost wall distances.
2. Independent ZPG-boundary-layer measurements and DNS recover wall-attached packet and velocity structures through the logarithmic layer, reaching approximately $0.2\delta_{99}$.
3. Interface-conditioned statistics include the local intermittent valleys and peaks and show that the normalised straining environment in the interface buffer is essentially the same as in the turbulent core.
4. Turbulent-level vorticity and strain then terminate sharply at the T/NT interface; the exterior branch is much weaker and potential.

The weak inference is that the attached gradient clusters known in channels are the carriers of the wall-similar physics observed in the boundary-layer core and remain its carriers up to the local interface, including canopy peaks above δ_{99} . On that basis, we hypothesise that the intermittent region is the sparse upper support of the wall-attached hierarchy and that the T/NT interface is its convoluted outer envelope. The inference is weak because the channel clusters, boundary-layer attached velocity structures and interface-conditioned gradients are different observables from different datasets; it is nevertheless constrained by their common scaling and dynamics.

A joint object/interface analysis provides a direct test: vorticity clusters, Q structures and the interface should be extracted simultaneously from the same TBL snapshots and their conditional intersections measured. The observation that more than 80% of channel vortex clusters intersect a Q object supplies a concrete expectation for that test.

A similarly careful statement is required for vorticity. For incompressible, constant-density flow,

$$\frac{D\boldsymbol{\omega}}{Dt} = (\boldsymbol{\omega} \cdot \nabla) \mathbf{u} + \nu \nabla^2 \boldsymbol{\omega}. \quad (8)$$

There is no baroclinic source in equation (8). The wall supplies vorticity, stretching and tilting redistribute and amplify it, and viscosity diffuses it across the T/NT interface. This global statement does not imply a Lagrangian conveyor belt in which identifiable vortex objects travel intact from the wall to δ_{99} . The tracking evidence argues against that literal interpretation.

6 Boundary layers and channels

A channel is not a boundary layer with a reflecting outer boundary. Jiménez et al. [2010] show that large-scale velocity correlations cross the channel centreline; ejections from one half can penetrate the other. A TBL replaces this centreline interaction with an intermittent mixture of rotational and potential fluid and with associated pressure fluctuations. Figure 9 demonstrates centreline coupling but does not demonstrate a destructive collision between two attached hierarchies.

Higher-Reynolds-number one-point statistics show that all three velocity-component intensities exceed their channel values in the TBL, with the largest difference near $y/\delta \simeq 0.35\text{--}0.5$ and little movement of that location over the available Reynolds-number range [Sillero et al., 2013]. This is consistent with an edge-mediated pressure and entrainment effect, not with a dynamically empty outer TBL.

Three-dimensional correlations provide the strongest correction to the earlier “channel traffic jam” interpretation [Sillero et al., 2014]. Weak streamwise-velocity correlations reach approximately 18δ in channels but only approximately 7δ in boundary layers. Their cross-flow organisation is nevertheless qualitatively similar. Figure 10 shows that the channel streamwise correlation length continues to grow to $y/\delta \simeq 0.5$, while the boundary-layer value levels off much earlier. The source study argues that intermittency prevents the corresponding global structure from forming in the TBL.

The defensible comparison is therefore:

- a channel has no T/NT interface, communicates across its centreline and supports very long global streamwise modes;
- a boundary layer has an open intermittent edge, stronger pressure and transverse-velocity fluctuations, shorter streamwise correlations and a weak potential-flow response beyond the vorticity boundary.

“Unconfined” does not mean dynamically simpler. It means that the outer boundary condition is an irrotational free stream rather than another turbulent half-channel.

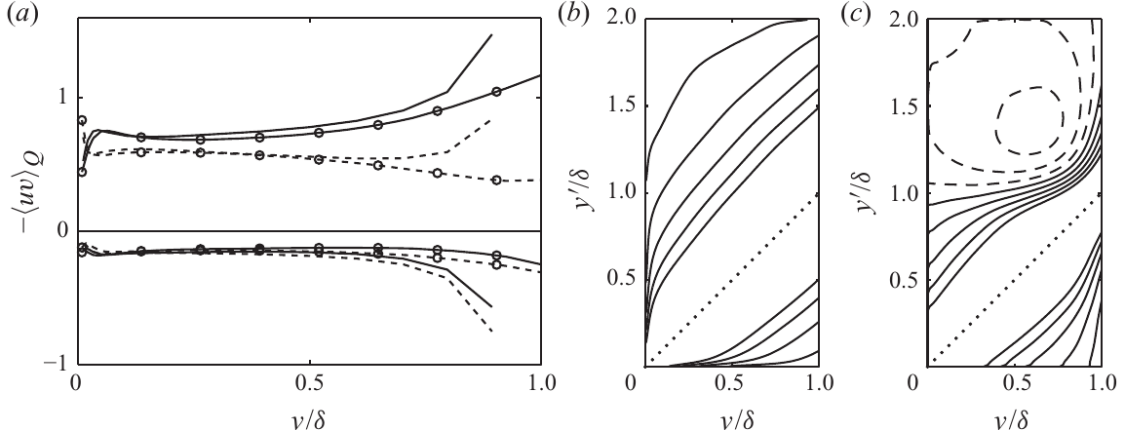


Figure 9: Quadrant contributions in a channel and a boundary layer (a), and wall-normal (b) and streamwise (c) large-scale channel correlations. Contours crossing $y/\delta = 1$ demonstrate centreline coupling. Reproduced from figure 5 of Jiménez et al. [2010].

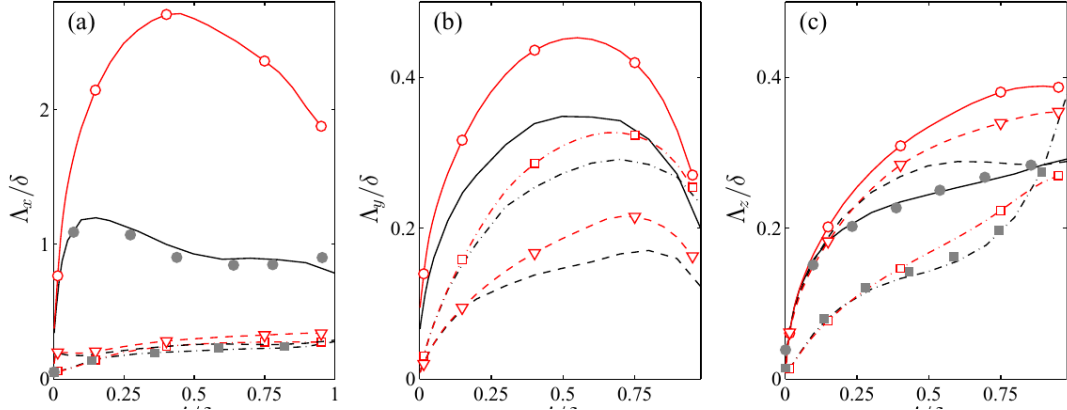


Figure 10: Integral correlation lengths for the three velocity components. In panel (a), the red channel curve for streamwise velocity continues to grow to $y/\delta \simeq 0.5$, whereas the black boundary-layer curve levels off earlier. Reproduced from figure 10 of Sillero et al. [2014].

7 Direct tests of the hypothesis

The canopy interpretation can be tested quantitatively using existing DNS fields or modest extensions of them.

7.1 Joint object/interface labelling

For each snapshot, identify the T/NT interface with the established convention of Borrell and Jiménez [2016], together with attached vortex clusters and attached Q objects. Measure

$$P(d_{\text{TNTI}} < a\lambda \mid \text{top of attached object}) \quad (9)$$

and compare it with randomised objects of the same size and height. A canopy interpretation predicts an excess near the tops of the largest objects.

7.2 Coverage rather than height alone

Construct the pointwise union of objects and compare its coverage

$$\gamma_Q(y; H) = \Pr\{(x, y, z) \in \cup_k Q_k(H)\} \quad (10)$$

with the vorticity intermittency in equation (4). Object footprints and overlaps must be retained; a cumulative height distribution is insufficient.

7.3 Conditional budgets at structural tips

Evaluate mean-shear production, nonlinear transport, vortex stretching and viscous diffusion conditioned jointly on distance from the interface and distance from an object top. If sparse object tips form the interface, these budgets should differ from those at generic interface points of the same height. The test would determine whether the core-like straining on the turbulent side in figure 8 is specifically associated with attached cluster tips.

7.4 Matched internal/external comparison

Use identical thresholds and object definitions at matched Re_τ . Compare object populations, height scaling, centreline or interface termination and canopy coverage. This would replace qualitative collision and freedom metaphors with measurable differences.

8 Conclusions

The published evidence supports a unified but qualified structural narrative.

1. **Hierarchy.** Three-dimensional vortex clusters and intense Reynolds-stress objects contain attached, approximately self-similar families whose wall-parallel dimensions scale with their height. At the threshold of [Lozano-Durán et al. \[2012\]](#), attached Q objects carry about 60% of the Reynolds stress while covering less than 8% of a plane.
2. **Wall similarity.** Tracked tall attached structures preserve geometrical and temporal similarity: their dimensions scale with height and their lifetimes with Δ_y/u_τ . Independent boundary-layer studies recover analogous attached organisation to approximately $0.2\delta_{99}$.
3. **Termination at the edge.** Turbulent-level vorticity and strain terminate near the interface in a drop of more than one order of magnitude. Core-like normalised straining persists up to the turbulent side, while only a weak exponentially decaying potential branch remains outside.
4. **Internal versus external flow.** Channel structures and correlations cross the centreline, whereas TBLs interact with potential fluid through intermittency and pressure. Channels possess longer, not shorter, global streamwise correlations.
5. **Weak cluster-to-interface connection.** Because the interface buffer has essentially the same normalised straining environment as the core, we infer that the wall-attached hierarchy carries its wall-similar physics from approximately $0.2\delta_{99}$ to the local interface, including peaks above δ_{99} . The T/NT interface is proposed to be the envelope where that sparse hierarchy terminates.

The main value of the canopy framework is not that it abolishes the outer layer. It proposes a weak, testable identification between the wall-similar attached hierarchy and the structural support of the T/NT interface. Figures 7 and 8 constrain that identification by showing respectively where turbulent-level gradients terminate and that their normalised organisation remains core-like up to the turbulent side.

Data availability

The DNS statistics and several datasets associated with the cited studies are available through the Fluid Dynamics Group at Universidad Politécnica de Madrid: <https://torroja.dmt.upm.es/>, <https://torroja.dmt.upm.es/turbdata/> and <https://torroja.dmt.upm.es/paperdat/>. Availability differs by paper.

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